

Tech & Teach

Digital Chip Design Program

RTL Design

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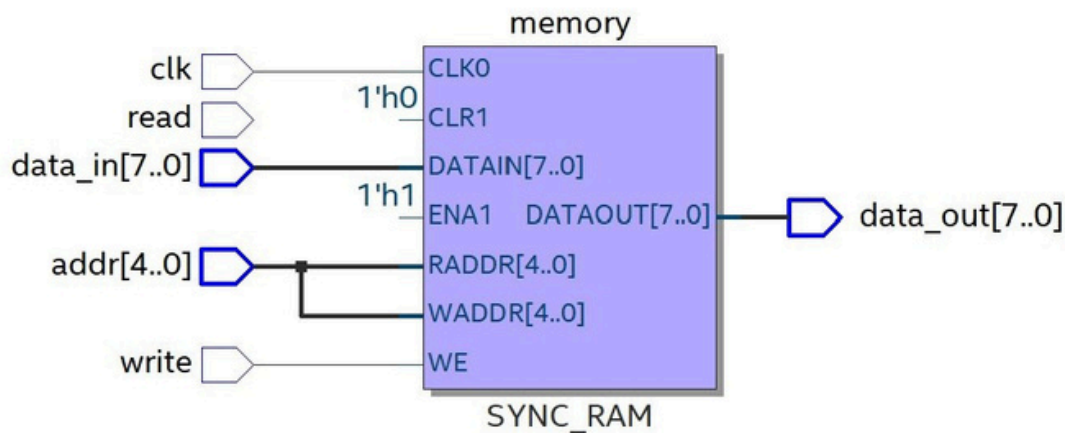
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Seq & Comb Circuits Datapath – Part 1

Version 1.0

Task #10.1

```
module mem #(parameter Width=8,Depth=32) (  
    input      clk,  
    input      read,  
    input      write,  
    input  logic [$clog2(Depth)-1:0] addr,  
    input  logic [Width-1:0] data_in,  
    output logic [Width-1:0] data_out  
);  
    logic [Width-1:0] memory [0:Depth-1];  
    always@(posedge clk) begin  
        if (write)  
            memory[addr] <= data_in;  
        end  
        assign data_out = memory[addr];  
    endmodule
```



Task #10.1

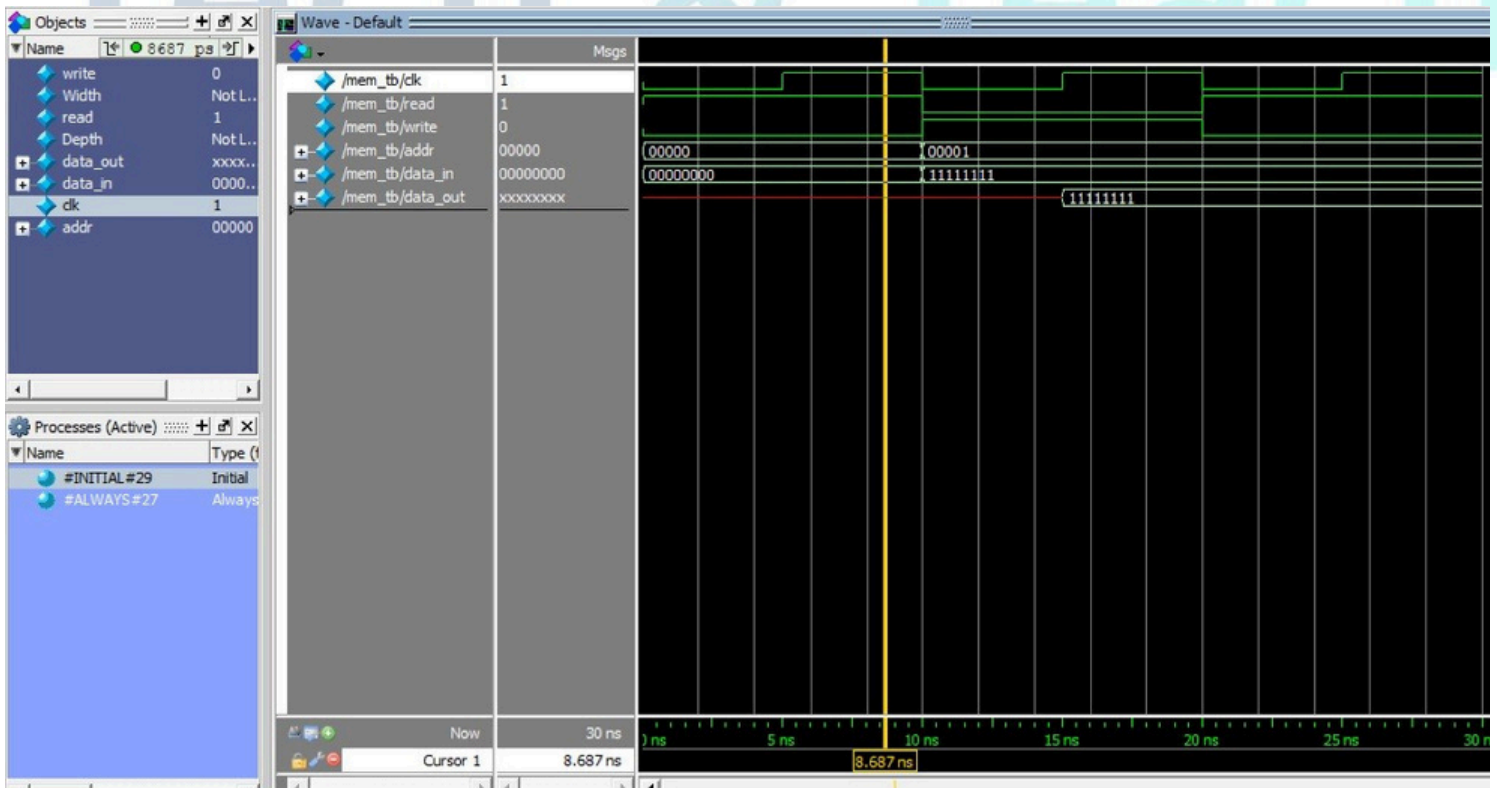
```
`timescale 1ns/1ps
module mem_tb;
parameter Width = 8;
parameter Depth = 32;
logic clk;
logic read;
logic write;
logic [$clog2(Depth)-1:0] addr;
logic [Width-1:0] data_in;
logic [Width-1:0] data_out;
mem #(
.Width(Width),
.Depth(Depth)
) dut (
.clk(clk),
.read(read),
.write(write),
.addr(addr),
.data_in(data_in),
.data_out(data_out)
);
```

```
always #5 clk = ~clk;
initial begin
clk = 0;
read = 1;
write = 0;
addr = 0;
data_in = 8'h00;

#10;
addr = 1;
read = 0;
write = 1;
data_in = 8'hFF;

#10;
write = 0;
read = 1;
addr = 1;

#10;
$finish;
end
endmodule
```



Work distribution

Section	Almas	Atheer	Mjd	Maha	Jood
Report	✓	✓	✓	✓	✓
Task9.1	✓	✓	✓	✓	✓
Task9.2	✓	✓	✓	✓	✓
Task9.3	✓	✓	✓	✓	✓

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